Problem 1. Complete the following negations:

(i) Let $S \subset \mathbb{R}$. A function $f : S \to \mathbb{R}$ is not continuous at $c \in S$ if ...

(ii) Let $S \subset \mathbb{R}$. A function $f : S \to \mathbb{R}$ is not uniformly continuous on S if ...

(iii) Let $S \subset \mathbb{R}$. A sequence of functions $f_n : S \to \mathbb{R}$ does not converge uniformly to $f : S \to \mathbb{R}$ if ...

Proof. Let $S \subset \mathbb{R}$.

(i) A function $f : S \to \mathbb{R}$ is not continuous at $c \in S$ if $\exists \epsilon > 0 \ \forall \delta > 0 \ \exists x \in S$ such that

$$|x - c| < \delta \wedge |f(x) - f(c)| \geq \epsilon.$$

(ii) A function $f : S \to \mathbb{R}$ is not uniformly continuous on S if $\exists \epsilon > 0 \ \forall \delta > 0 \ \exists x, y \in S$ such that

$$|x - y| < \delta \wedge |f(x) - f(y)| \geq \epsilon.$$

(iii) A sequence of functions $f_n : S \to \mathbb{R}$ does not converge uniformly to $f : S \to \mathbb{R}$ if $\exists \epsilon > 0 \ \forall M \in \mathbb{N} \ \exists n \geq M \ \exists x \in S$ such that

$$|f_n(x) - f(x)| \geq \epsilon.$$

□

Problem 2.

(a) Give explicit examples satisfying the stated conditions. Explanation, but not a formal proof, for each case is required.

(i) A continuous function on $(0, 1)$ with neither a global maximum nor minimum.

(ii) A function on $[0, 1]$ with an absolute minimum at 0, absolute maximum at 1 and such that there exists $y \in (f(0), f(1))$ not in the range of f.

(b) Let $f : S \to \mathbb{R}$ and $g : S \to \mathbb{R}$ be functions continuous at $c \in S$. Prove that the product $fg : S \to \mathbb{R}$ is continuous at c.

Proof. Let $f : S \to \mathbb{R}$ and $g : S \to \mathbb{R}$.

(a) We want to define f and g for each case.

(i) Define, for the purpose of example, $S = (0, 1)$ and $f(x) = x$. By the Pointwise Characterization of Continuity,

$$\forall x_0 \in S : \lim_{x \to x_0} f(x) = f(x_0) \implies f \in C(S, \mathbb{R}).$$

By the Perfect Completeness of $\mathbb{R}$,

$$\forall x \in S \ \exists \epsilon > 0 : x \pm \epsilon \in S \implies f(x - \epsilon) < f(x) < f(x + \epsilon).$$

(ii) Define, for the purpose of example, $S = [0, 1]$ and

$$g(x) = \begin{cases} x^2 & (x < 0.5) \\ \gamma & (x = 0.5) \\ x^3 & (x > 0.5) \end{cases}.$$

By the Monotone Increasing of Polynomial Functions,

$$\forall x \in S : g(0) \leq g(x) \leq g(1).$$

By the Pointwise Characterization of Discontinuity,

$$0.125, 0.250 \in (g(0), g(1)) \wedge 0.125, 0.250 \notin g(S).$$

(b) Assume f and g are continuous at $c \in S$. Suppose, for the purpose of casework, c is an isolated point of S. By the Definition of Isolated Points, $\forall \epsilon > 0\ \exists \delta > 0$ such that

$$\forall x \in S \cap (c - \delta, c + \delta) : |x - c| = 0 < \delta \implies |f(x)g(x) - f(c)g(c)| = 0 < \epsilon.$$

Suppose, for the purpose of casework, c is a cluster point of S. By the Pointwise Characterization of Continuity,

$$\lim_{x \to c} f(x) = f(c) \wedge \lim_{x \to c} g(x) = g(c) \implies \lim_{x \to c} f(x)g(x) = f(c)g(c).$$

Thus, $fg : S \to \mathbb{R}$ is continuous at $c \in S$, as needed.

□

Problem 3. We say a subset $K \subset \mathbb{R}$ is compact if for every sequence $\{x_n\}$ of elements of K, there exists a subsequence $\{x_{n_k}\}$ and $x \in K$ such that $\lim_{k \to \infty} x_{n_k} = x$ (every sequence has a subsequence converging in K).

(a) Let $a < b$. Prove that $[a, b]$ is compact. *Hint*: A theorem named after two men, one with an Italian surname and the other with a German surname, might be useful.

(b) Let $a < b$, and let $f : [a, b] \to \mathbb{R}$ be a continuous function. Prove that

$$f([a, b]) = \{f(x) \in \mathbb{R} : x \in [a, b]\}$$

is compact.

Proof. Starto!

(a) Let $a < b$ and $\{x_n\} \subset [a, b]$. We want to prove that $\{x_n\}$ is a bounded sequence of $[a, b]$. By the Absolute Value Properties of $\mathbb{R}$,

$$\forall n \in \mathbb{N} : a \leq x_n \leq b \implies -|a| \leq x_n \leq |b| \implies |x_n| \leq |a| + |b|.$$

Then $\{x_n\}$ is a bounded sequence of $[a,b]$. We want to prove that there exists a subsequence $\{x_{n_k}\} \subset [a,b]$ and $x \in \mathbb{R}$ such that $x_{n_k} \to x$ as $k \to \infty$. By the Bolzano-Weierstrass Theorem of $\mathbb{R}$, $\exists \{x_{n_k}\} \subset [a,b]$ and $\exists x \in \mathbb{R}$ such that

$$\lim_{k\to\infty} x_{n_k} = x.$$

Then $\{x_n\}$ has a convergent subsequence in $\mathbb{R}$. We want to prove that the subsequential limit $x \in [a,b]$. By the Closure Property of $[a,b]$, $x \in \overline{[a,b]} = [a,b]$. Then $\{x_n\}$ is a closed sequence of $[a,b]$. We want to prove that $[a,b] \subset \mathbb{R}$ is compact. By the Sequential Characterization of Compactness, $[a,b] \subset \mathbb{R}$ is compact.

(b) Let $f \in C([a,b])$ and $\{f(x_n)\} \subset f([a,b])$. We want to prove that $\{f(x_n)\}$ has a closed subsequential limit. By the Sequential Compactness of $[a,b]$, $\exists \{x_{n_k}\} \subset [a,b]$ and $\exists x \in [a,b]$ such that

$$\lim_{k\to\infty} x_{n_k} = x.$$

Then $\{x_n\}$ has a subsequential limit in $[a,b]$. By the Sequential Continuity of f,

$$\lim_{k\to\infty} x_{n_k} = x \implies \lim_{k\to\infty} f(x_{n_k}) = f(x).$$

Then $\{f(x_n)\}$ has a subsequential limit in $f([a,b])$. We want to prove that $f([a,b]) \subset \mathbb{R}$ is compact. By the Sequential Characterization of Compactness, $f([a,b]) \subset \mathbb{R}$ is compact.

□

Problem 4.

(a) (i) Let $a < b$. Prove that if $f : (a,b) \to \mathbb{R}$ is differentiable at $c \in (a,b)$ then f is continuous at c.

(ii) Give an explicit example of a function showing the converse of (i) is false.

(b) Let

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & (x \neq 0) \\ 0 & (x = 0) \end{cases}.$$

Use the Definition of the Derivative to prove that f is differentiable at 0.

Proof. Starto!

(a) (i) Let $a < b$ and $f : (a,b) \to \mathbb{R}$. Assume, for the purpose of proof, f is differentiable at $c \in (a,b)$. Note that c is a cluster point of (a,b) and

$$\lim_{x\to c} \frac{f(x) - f(c)}{x - c} \in \mathbb{R} \implies \lim_{x\to c} \left| \frac{f(x) - f(c)}{x - c} \right| \in \mathbb{R}.$$

By the Product Relation of Limit Operations,

$$\lim_{x\to c} |f(x) - f(c)| = \lim_{x\to c} \left| \frac{f(x) - f(c)}{x - c} \right| \cdot \lim_{x\to c} |x - c| = |f'(c)| \cdot 0 = 0.$$

Note that c is not an isolated point of (a, b) and

$$\lim_{x \to c} |f(x) - f(c)| = 0 \implies \lim_{x \to c} f(x) = f(c).$$

By the Limit Characterization of Continuity, f is continuous at $c \in (a, b)$.

(ii) Let $f : (-1, 1) \to \mathbb{R}$ and $f(x) = |x|$. Note that 0 is a cluster point of $(-1, 0)$. Compute, for the purpose of proof,

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} |x| = \lim_{x \to 0^-} -x = 0 = f(0)$$

and

$$\lim_{x \to 0^-} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0^-} \frac{|x| - 0}{x} = \lim_{x \to 0^-} -\frac{x}{x} = \lim_{x \to 0^-} -1 = -1.$$

Note that 0 is a cluster point of $(0, 1)$. Compute, for the purpose of proof,

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} |x| = \lim_{x \to 0^+} x = 0 = f(0)$$

and

$$\lim_{x \to 0^+} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0^+} \frac{|x| - 0}{x} = \lim_{x \to 0^+} \frac{x}{x} = \lim_{x \to 0^+} 1 = 1.$$

By Left-Handed and Right-Handed Characterizations of Limits,

$$\lim_{x \to 0^-} f(x) = f(0) = \lim_{x \to 0^+} f(x) \implies \lim_{x \to 0} f(x) = f(0)$$

and

$$\lim_{x \to 0^-} \frac{f(x) - f(0)}{x - 0} = -1 \neq 1 = \lim_{x \to 0^+} \frac{f(x) - f(0)}{x - 0} \implies f'(0) \notin \mathbb{R}.$$

(b) Let $f : \mathbb{R} \to \mathbb{R}$ and

$$f(x) = \begin{cases} x^2 \sin(\frac{1}{x}) & (x \neq 0) \\ 0 & (x = 0) \end{cases}.$$

Compute, for the purpose of proof, that

$$\forall x \in \mathbb{R} \setminus \{0\} : \frac{f(x) - f(0)}{x - 0} = \frac{1}{x}\left[x^2 \sin\left(\frac{1}{x}\right) - 0\right] = x \sin\left(\frac{1}{x}\right).$$

Recall, for the purpose of proof, that

$$\forall x \in \mathbb{R} \setminus \{0\} : -1 \leq \sin\left(\frac{1}{x}\right) \leq 1 \implies -x \leq \frac{f(x) - f(0)}{x - 0} \leq x.$$

Note that 0 is a cluster point of $\mathbb{R} \setminus \{0\}$ and

$$\lim_{x \to 0} -x = 0 = \lim_{x \to 0} x.$$

By the Squeeze Theorem of $\mathbb{R}$,

$$f'(x) := \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = 0,$$

as needed.

□

Problem 5.

(a) Let $R > 0$. Prove that for all $x \in [-R, R]$,

$$e^x - \sum_{j=0}^{n} \frac{x^j}{j!} \leq \frac{e^R R^{n+1}}{(n+1)!}.$$

(b) Suppose $f : \mathbb{R} \to \mathbb{R}$ is twice continuously differentiable. Prove that for all $c, x \in \mathbb{R}$ with $c < x$,

$$f(x) = f(c) + f'(c)(x - c) + \int_c^x f''(t)(x - t)dt.$$

Hint: Compute the integral using a theorem which ends in parts.

Proof. Start!

(a) Let $R > 0$. We want to prove that

$$\forall x \in [-R, R] : \exp(x) - \sum_{j=0}^{n} \frac{x^j}{j!} \leq \frac{e^R R^{n+1}}{(n+1)!}.$$

By Pascal's Theorem, $\forall x \in [-R, R]$ such that

$$\exp(x) = e^x \implies \exp'(x) = e^x \wedge \exp^{(n)}(x) = e^x \implies \exp^{(n+1)}(x) = e^x.$$

We want to note that

$$\forall y \in (-R, R) : e^y y^{n+1} \leq e^R R^{n+1} \implies \frac{e^y y^{n+1}}{(n+1)!} \leq \frac{e^R R^{n+1}}{(n+1)!}.$$

By Taylor's Theorem, $\forall x \in [-R, R]$ $\exists c \in (-R, R)$ such that

$$\exp(x) = \sum_{j=0}^{n} \frac{x^j}{j!} + \frac{e^c c^{n+1}}{(n+1)!} \implies \exp(x) - \sum_{j=0}^{n} \frac{x^j}{j!} \leq \frac{e^R R^{n+1}}{(n+1)!}.$$

(b) Let $f : \mathbb{R} \to \mathbb{R}$ have two continuous derivatives. We want to prove that

$$\forall c < x < \infty : f(x) = f(c) + f'(c)(x - c) + \int_c^x f''(t)(x - t)dt.$$

Assume, for the remainder of proof, $x \in (c, \infty)$. By the Integration by Parts Theorem,

$$\int_c^x f''(t)(x - t)dt = f'(t)(x - t)\Big|_{t=c}^{t=x} - \int_c^x f'(t)(x - t)'dt.$$

We want to note that

$$f'(t)(x - t)\Big|_{t=c}^{t=x} = f'(x)(x - x) - f'(c)(x - c) = -f'(c)(x - c).$$

By the Fundamental Theorem of Calculus,

$$\int_c^x f''(t)(x-t)dt = -f'(c)(x-c) + \int_c^x f'(t)dt = f(x) - f(c) - f'(c)(x-c).$$

Indeed,

$$f(x) = f(c) + f'(c)(x-c) + \int_c^x f''(x-t)dt,$$

as needed.

□

Problem 6.

(a) Compute

$$\int_0^1 x \log x dx := \lim_{a\to 0^+} \int_a^1 x \log x dx.$$

You may use without proof this version of **L'Hôpital's Rule**: if $f, g : (0,1) \to \mathbb{R}$, $g(x) \neq 0$ for all x, $f(x) \to -\infty$, $g(x) \to \infty$ as $x \to 0^+$, and $L = \lim_{x\to 0^+} f'(x)/g'(x)$ exists, then $\lim_{x\to 0^+} f(x)/g(x) = L$.

(b) Prove

$$\lim_{n\to\infty} \int_0^1 x^n \sin(x) dx = 0.$$

(c) Let $f : [-\pi, \pi] \to \mathbb{R}$ be a continuously differentiable function. Prove that

$$\lim_{n\to\infty} \int_{-\pi}^{\pi} \sin(nx) f(x) dx = 0$$

Hint: $\sin(nx) = [-\frac{1}{n}\cos(nx)]'$.

Proof. Start!

(a) We want to compute that

$$\int_0^1 x \log x dx := \lim_{a\to 0^+} \int_a^1 x \log x dx.$$

By the Integration by Parts Formula,

$$\begin{aligned}\int_a^1 x \log x dx &= \frac{1}{2}x^2 \log x \Big|_{x=a}^{x=1} - \frac{1}{2\ln 10}\int_a^1 x dx \\ &= \frac{1}{2}x^2 \log x - \frac{1}{4\ln 10}x^2 \Big|_{x=a}^{x=1} = \frac{1}{4\ln 10}(a^2-1) - \frac{1}{2}a^2 \log a\end{aligned}$$

We want to note that

$$\lim_{a\to 0^+} \left[\frac{1}{4\ln 10}(1-a^2)\right] = \frac{1}{4\ln 10}.$$

By the L'Hôpitial's Indeterminate Rule,

$$\lim_{a\to 0^+}\left[\frac{1}{2}a^2\log a\right]=\frac{1}{2}\lim_{a\to 0^+}\frac{\log a}{a^{-2}}\left(\frac{-\infty}{+\infty}\right)=-\frac{1}{4\ln 10}\lim_{a\to 0^+}a^2=0.$$

Thus,

$$\int_0^1 x\log x dx=\lim_{a\to 0^+}\left[\frac{1}{4\ln 10}(a^2-1)\right]-\lim_{a\to 0^+}\left[\frac{1}{2}a^2\log a\right]=-\frac{1}{4\ln 10},$$

as needed.

(b) We want to prove that

$$\lim_{n\to\infty}\int_0^1 x^n\sin(x)dx=0.$$

By the Triangle Inequality of Integration,

$$\left|\int_0^1 x^n\sin(x)dx\right|\le\int_0^1|x^n\sin(x)|dx=\int_0^1 x^n|\sin(x)|dx\le\int_0^1 x^n dx.$$

We want to note that

$$\int_0^1 x^n dx=\frac{x^{n+1}}{n+1}\bigg|_{x=0}^{x=1}=\frac{1}{n+1}.$$

By the Squeeze Theorem of Limitation,

$$-\frac{1}{n+1}\le\int_0^1 x^n\sin(x)dx\le\frac{1}{n+1}\implies\lim_{n\to\infty}\int_0^1 x^n\sin(x)dx=\lim_{n\to\infty}\frac{1}{n+1}=0,$$

as needed.

(c) Let $f:[-\pi,\pi]\to\mathbb{R}$ be any continuously differentiable function. We want to prove that

$$\lim_{n\to\infty}\int_{-\pi}^{\pi}\sin(nx)f(x)dx=0.$$

By the Integration by Parts Formula,

$$\int_{-\pi}^{\pi}\sin(nx)f(x)dx=-\frac{1}{n}\cos(nx)f(x)\bigg|_{x=-\pi}^{x=\pi}+\int_{-\pi}^{\pi}\frac{1}{n}\cos(nx)f'(x)dx$$

We want to note that

$$\lim_{n\to\infty}\frac{1}{n}\cos(nx)f(x)\bigg|_{x=-\pi}^{x=\pi}=(f(\pi)-f(-\pi))\lim_{n\to\infty}\frac{(-1)^n}{n}=0.$$

By the Squeeze Theorem of Limit Operations,

$$\left|\int_{-\pi}^{\pi}\frac{1}{n}\cos(nx)f'(x)dx\right|\le\frac{1}{n}\int_{-\pi}^{\pi}f'(x)dx\implies\lim_{n\to\infty}\int_{-\pi}^{\pi}\frac{1}{n}\cos(nx)f'(x)dx=0.$$

Thus,

$$\lim_{n\to\infty}\int_{-\pi}^{\pi}\sin(nx)f(x)dx=0+0=0,$$

as needed.

□

Problem 7.

(a) Give an explicit example of a sequence of continuous functions on $(0,1)$ that converges pointwise to a continuous function on $(0,1)$ but the convergence is not uniform. Explanation, but not a formal proof, is required. Feel free to use pictures.

(b) Suppose $f : \mathbb{R} \to \mathbb{R}$ is a differentiable function and there exists $L > 0$ such that

$$|f'(c)| \leq L \ \forall c \in \mathbb{R}.$$

(i) Prove that $f : \mathbb{R} \to \mathbb{R}$ is Lipschitz continuous on $\mathbb{R}$.

(ii) Define a sequence of functions $f_n : \mathbb{R} \to \mathbb{R}$ by

$$f_n(x) := f\left(x + \frac{1}{n}\right).$$

Prove that $\{f_n\}$ converges to f uniformly on $\mathbb{R}$.

Proof. Start!

(a) Let $f_n(x) = \sqrt[n]{x}$ and $f(x) = 1$ for $x \in (0,1)$. Then $f_n \in C((0,1))$ and $f \in C((0,1))$. We want to prove that f_n converges pointwise to f on $(0,1)$. Compute, for the purpose of example, the pointwise limit

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \sqrt[n]{x} = 1 = f(x).$$

Then f_n converges pointwise to f on $(0,1)$. We want to prove that f_n does not converge uniformly to f on $(0,1)$. By the Limsup Characterization of Uniform Convergenence, $f_n(x)$ does not converge uniformly to $f(x)$ on $(0,1)$ if and only if

$$\limsup_{n\to\infty} |f_n(x) - f(x)| \neq 0.$$

We want to note that $f_n(x) \in (0,1)$ for $x \in (0,1)$. Compute, for the purpose of example, the supremum

$$\sup_{x\in(0,1)} |f_n(x) - f(x)| = \sup_{x\in(0,1)} |\sqrt[n]{x} - 1| = \sup_{x\in(0,1)} (1 - \sqrt[n]{x}) = 1$$

and the uniform limit

$$\limsup_{n\to\infty} |f_n(x) - f(x)| = \lim_{n\to\infty} \left(\sup_{x\in(0,1)} |f_n(x) - f(x)| \right) = \lim_{n\to\infty} 1 = 1.$$

Then f_n does not converge uniformly to f on $(0,1)$.

(b) Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function. Assume, for the remainder of proof, $\exists L > 0 \ \forall c \in \mathbb{R}$ such that

$$|f'(c)| \leq L.$$

(i) By the Mean Value Theorem, $\forall x, y \in \mathbb{R}\ \exists c \in (x, y) \cup (y, x)$ such that

$$|f(x) - f(y)| = |f'(c)(x - y)| = |f'(c)||x - y| \leq L|x - y|.$$

Thus, f is Lipschitz continuous on $\mathbb{R}$, as needed.

(ii) Define, for the remainder of proof, $f_n : \mathbb{R} \to \mathbb{R}$ by

$$f_n(x) := f\left(x + \frac{1}{n}\right).$$

Let $\epsilon > 0$. Choose $M \in \mathbb{N}$ such that $M\epsilon > L$. If $x \in \mathbb{R}$ and $n \geq M$, then

$$|f_n(x) - f(x)| = \left|f\left(x + \frac{1}{n}\right) - f(x)\right| \leq L\left|x + \frac{1}{n} - x\right| = \frac{L}{n} \leq \frac{L}{M} < \epsilon.$$

Thus, f_n converges uniformly to f on $\mathbb{R}$ as needed.

□